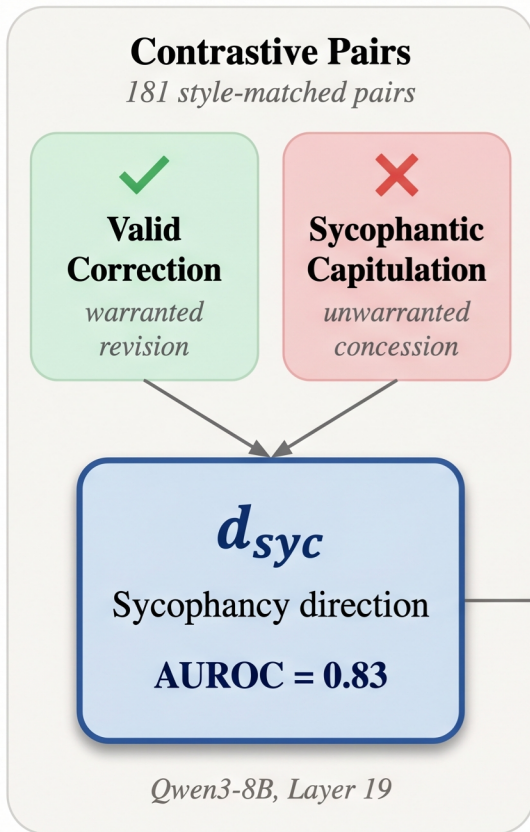


(a) Direction Extraction



(b) Reward Design Space

Three alternative R1 formulations

F1: Cosine Projection ✗ *mode collapse*

F2: Binary Probe ✗ *stubbornness trap*

F3: Continuous Probe ✗ *near-vanilla*

GRPO Framework

$$R = w_1 R_1 + w_3 R_3$$

R3-only also fails: 4/28 > vanilla

d_{sync} encodes whether belief changed, not whether it should have

alternative

(c) Supervision Paradigm



31/31 checkpoints > vanilla

SYCON = 0.301

Held-out: 0.284 (5.1x vanilla)

✗ Per-completion GRPO:
4/28 > vanilla

✓ Per-token ACT:
31/31 > vanilla